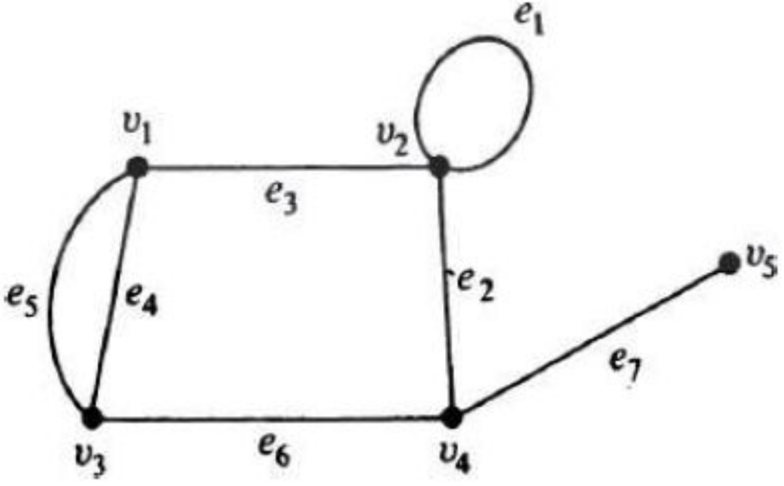
Introduction

WHAT IS A GRAPH?

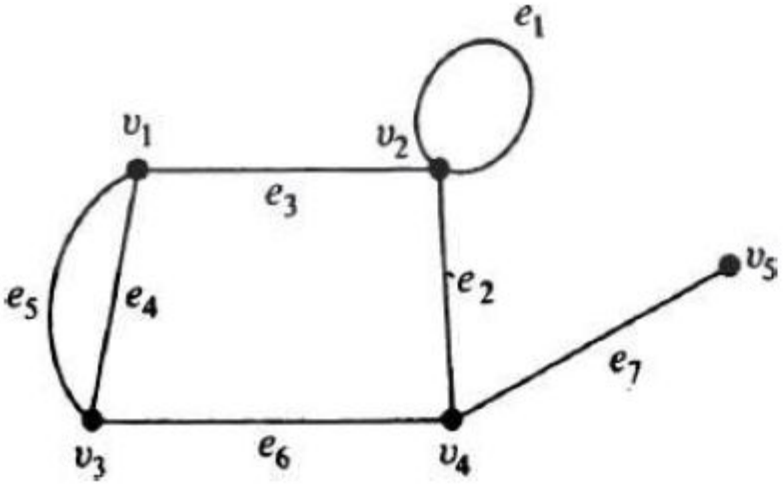
A linear graph (or simply a graph) $G = (V, E)$ consists of a set of objects $V = \{v_1, v_2, \ldots\}$ called vertices, and another set $E = \{e_1, e_2, \ldots\}$, whose elements are called edges, such that each edge e_k is identified with an unordered pair (v_i, v_j) of vertices



SELF-LOOP

Observe that this definition permits an edge to be associated with a same vertex pair (v_i, v_i). Such an edge having the same vertex as both its end vertices is called a **self-loop**.

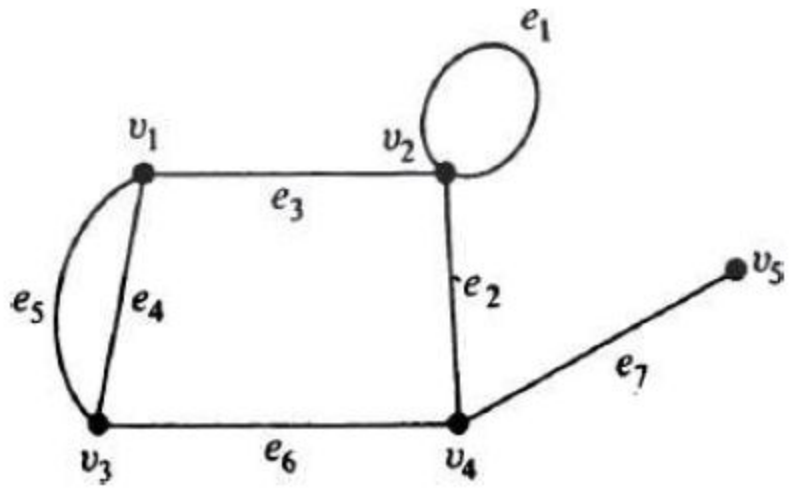
Here, e_1 is a self-loop.



PARALLEL EDGE

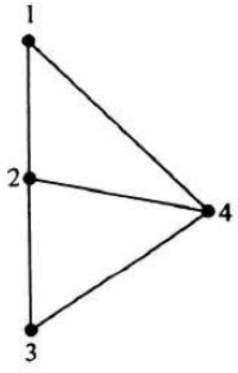
Also note that the definition allows more than one edge associated with a given pair of vertices. Such edges are referred to as **parallel edges**.

Here, e_4 and e_5 are parallel edges.

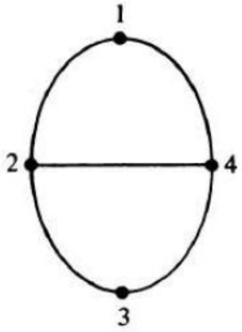


SIMPLE GRAPH

- A graph that has neither self-loops nor parallel edges is called a simple graph. In some graph-theory literature, a graph is defined to be only a **simple graph**.
- It should also be noted that, in drawing a graph, it is immaterial whether the lines are drawn straight or curved, long or short: what is important is the incidence between the edges and vertices.



(a)

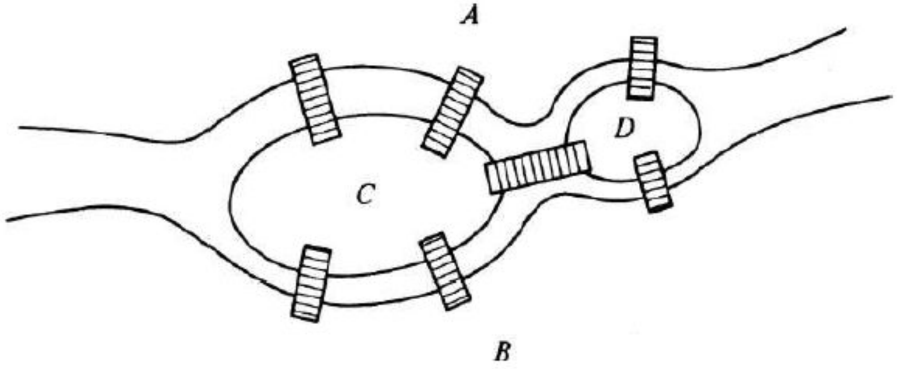


(b)

APPLICATIONS OF GRAPHS

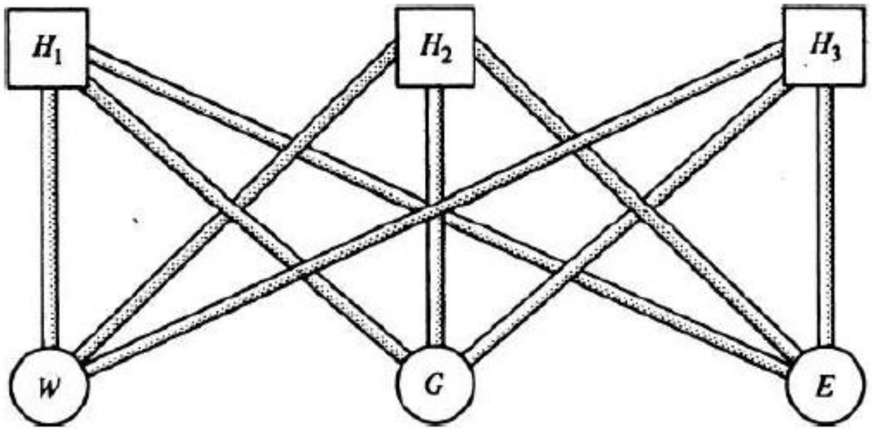
A graph can be used to represent almost any physical situation involving discrete objects and a relationship among them.

- Königsberg Bridge Problem: The problem is to start at any of the four land areas of the city, A, B, C, or D, walk over each of the seven bridges exactly once, and return to the starting point (without swimming across the river, of course).



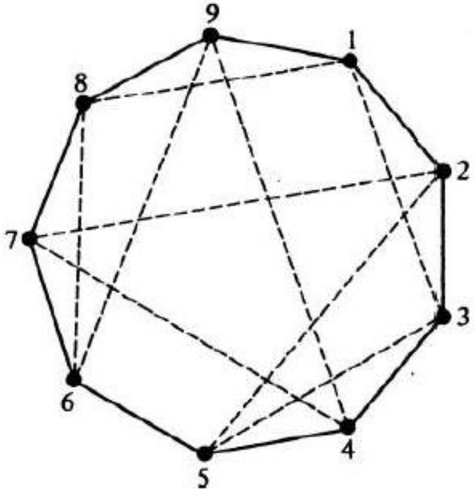
APPLICATIONS OF GRAPHS

- Utilities Problem: There are three houses H1, H2, and H3, each to be connected to each of the 3 utilities — water (W), gas (G), and electricity (E) by means of conduits. Is it possible to make such connections without any crossovers of the conduits?



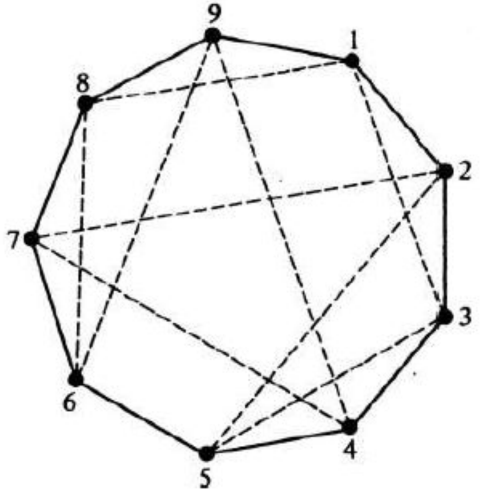
APPLICATIONS OF GRAPHS

- Seating Problem: Nine members of a new club meet each day for lunch at a round table. They decide to sit such that every member has different neighbors at each lunch. How many days can this arrangement last?



APPLICATIONS OF GRAPHS

- Seating Problem: The figure shows two possible seating arrangements. These are 1 2 3 4 5 6 7 8 9 1 (solid lines), and 1 3 5 2 7 4 9 6 8 1 (dashed lines).
  - It can be shown by graph-theoretic considerations that there are only two more arrangements possible. They are 1 5 7 3 9 2 8 4 6 1 and 1 7 9 5 8 3 6 2 4 1.
- In general it can be shown that for n people the number of such possible arrangements is



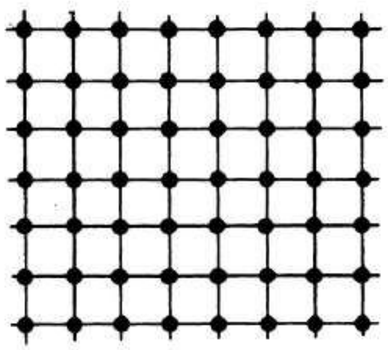
$$\frac{n-1}{2}, \quad \text{if } n \text{ is odd,}$$

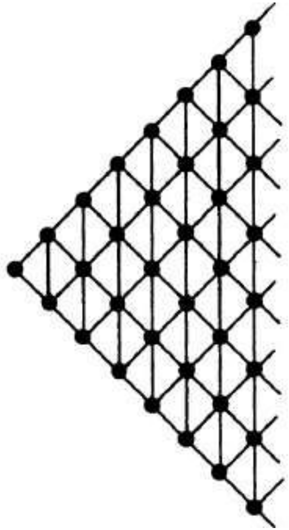
$$\frac{n-2}{2}, \quad \text{if } n \text{ is even.}$$

FINITE AND INFINITE GRAPHS

A graph with a finite number of vertices as well as a finite number of edges is called a **finite graph**; otherwise, it is an **infinite graph**.

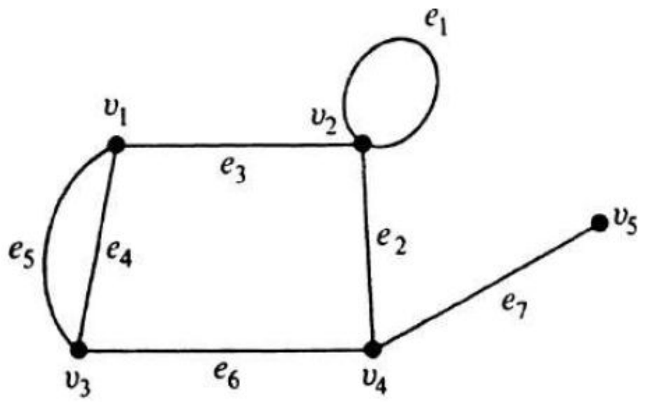
For this course, we shall confine ourselves to the study of finite graphs, and unless otherwise stated the term “graph” will always mean a finite graph.





INCIDENCE AND DEGREE

- When a vertex v_i is an end vertex of some edge e_j, v_i and e_j are said to be incident with (on or to) each other. In the figure, edges e_2, e_6, and e_7 are incident with vertex v_4.
- The number of edges incident on a vertex v_i, with self-loops counted twice, is called the degree, $d(v_i)$ of vertex v_i.
  - In the figure, $d(v_1) = d(v_3) = d(v_4) = 3$, $d(v_2) = 4$, and $d(v_5) = 1$.
  - The degree of a vertex is sometimes also referred to as its valency.



THEOREM 1-1

"The number of vertices of odd degree in a graph is always even."

For proving this theorem, we will start by stating an obvious observation.

Since each edge contributes two degrees, the sum of the degrees of all vertices in graph G is twice the number of edges in G. That is,

$$\sum_{i=1}^{n} d(v_i) = 2e. \tag{1-1}$$

THEOREM 1-1

"The number of vertices of odd degree in a graph is always even."

Now, the actual proof can be given as follows:

If we consider the vertices with odd and even degrees separately, the quantity in the left side of Eq. (1-1) can be expressed as the sum of two sums, each taken over vertices of even and odd degrees, respectively, as follows:

$$\sum_{i=1}^{n} d(v_i) = \sum_{\text{even}} d(v_j) + \sum_{\text{odd}} d(v_k). \qquad (1\text{-}2)$$

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Since the left-hand side in Eq. (1-2) is even, and the first expression on the right-hand side is even (being a sum of even numbers), the second expression must be

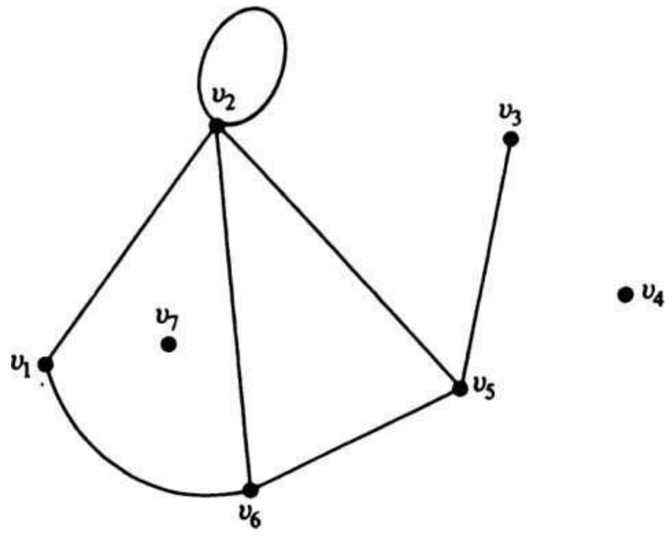
$$\sum_{\text{odd}} d(v_k) = \text{an even number.} \qquad (1\text{-}3)$$

Because in Eq. (1-3) each $d(v_k)$ is odd, the total number of terms in the sum must be even to make the sum an even number. Hence the theorem

ISOLATED VERTEX and PENDANT VERTEX

A vertex having no incident edge is called an isolated vertex. In other words, isolated vertices are vertices with zero degree. Vertices v_4 and v_7 in Fig. 1-11, for example, are **isolated vertices**.

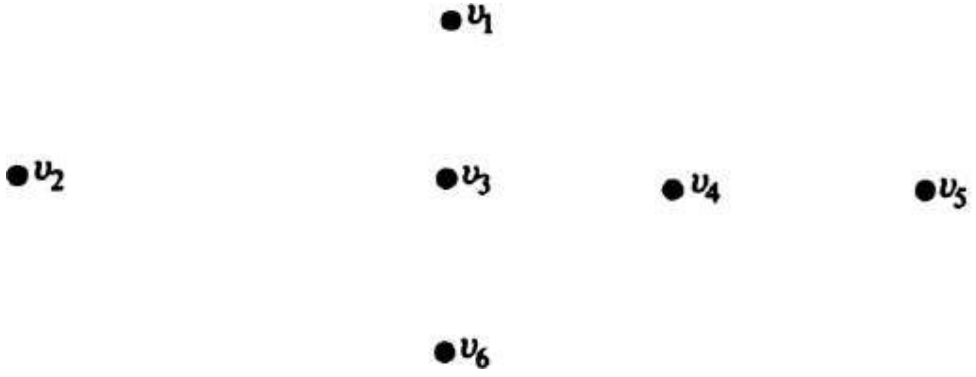
A vertex of degree one is called a **pendant vertex** or an end vertex. Vertex v_3 in Fig. 1-11 is a pendant vertex. Two adjacent edges are said to be in series if their common vertex is of degree two. In Fig. 1-11, the two edges incident on v_1 are in **series**.



NULL GRAPH

In the definition of a graph G = (V, E), it is possible for the edge set E to be empty. Such a graph, without any edges, is called a **null graph**. In other words, every vertex in a null graph is an isolated vertex.

Although the edge set E may be empty, the vertex set V must not be empty; otherwise, there is no graph. In other words, by definition, a graph must have at least one vertex.



BRIEF HISTORY OF GRAPH THEORY

- Graph theory was born in 1736 with Euler's paper in which he solved the Königsberg bridge problem
- In 1847, G. R. Kirchhoff (1824-1887) developed the theory of trees for their applications in electrical networks. Ten years later, A. Cayley (1821-1895) discovered trees while he was trying to enumerate the isomers of saturated hydrocarbons $C_nH_{2n + 2}$
- About the time of Kirchhoff and Cayley, two other milestones in graph theory were laid. One was the **four-color conjecture**, which states that four colors are sufficient for coloring any atlas (a map on a plane) such that the countries with common boundaries have different colors.
- The other milestone is due to Sir W. R. Hamilton (1805-1865). He invented a puzzle involving a 20 corner dodecahedron (where each corner represented a city). The object in the puzzle was to find. a route along the edges of the dodecahedron, passing through each of the 20 cities exactly once.